

Sheet 1

CS402/MA492 Cryptography, Tobias Rossmann, Semester 2, 2019/2020

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The Python module `cs402` (see <http://www.maths.nuigalway.ie/~rossmann/cs402/cs402.py>) contains functionality that you may find useful while working on this problem sheet.

Links on this sheet are clickable in the PDF version (but not in the HTML version). All files are also available on the course web page.

The problems below use the following alphabets:

27-letter alphabet

ABCDEFGHIJKLMNOPQRSTUVWXYZ_

in which the final symbol `_` indicates a space.

68-symbol alphabet

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789.,;:_\n

in which `_` indicates a space and `\n` indicates a newline character.

1	Answer the following questions. You will receive 1 mark for each correct answer, -1 marks for each incorrect answer, and 0 marks for each question that you choose not to answer. The minimal total number of marks is zero.	
	Cryptography is the science of secret writing.	☐ true / ☐ false
	$164^{29} \bmod 7 = 4$	☐ true / ☐ false
	The main objective of cryptanalysis is to devise efficient encryption algorithms.	☐ true / ☐ false
	Shift ciphers are impervious against frequency analysis.	☐ true / ☐ false
	Affine ciphers are special cases of shift ciphers.	☐ true / ☐ false
2	A shift cipher over the above 68-symbol alphabet was used to produce the ciphertext in the file http://www.maths.nuigalway.ie/~rossmann/cs402/a1q2-cipher6.txt We know that the plaintext consists of text written in English. Find the encryption key that was used to produce the ciphertext. Then use the same cipher with the same key to encrypt the string “Gulliver” of length 8. Carefully enter the resulting ciphertext as a string of precisely(!) 8 symbols; note that capitalisation matters.	
3	An affine cryptosystem on single letter message units over the above 68-symbol alphabet was used to produce the ciphertext in the file http://www.maths.nuigalway.ie/~rossmann/cs402/a1q3-cipher2.txt Find the encryption key that was used to produce the ciphertext. Then use the same cipher with the same encryption key to encrypt the string “mathematical” of length 12. Carefully enter the resulting ciphertext as a string of precisely 12 symbols.	

4	A Vigenère cipher over the above 27-symbol alphabet was used to produce the ciphertext in the file http://www.maths.nuigalway.ie/~rossmann/cs402/a1q4-cipher1.txt. The plaintext begins: I HAD BETTER SAY SOMETHING HERE ABOUT THIS Find the encryption key that was used to produce the ciphertext. Then use the same Vigenère cipher with the same encryption key to encrypt the string ‘MATHEMATICAL’ of length 12. Carefully enter the resulting ciphertext as a string of precisely 12 symbols.	
You can resubmit your answers any time up to the deadline. Only the most recent submission prior to the deadline will be marked. Each of the four problems above is worth a maximum of five marks. Incorrect answers to Problems 2–4 will be awarded 0/5 marks.		